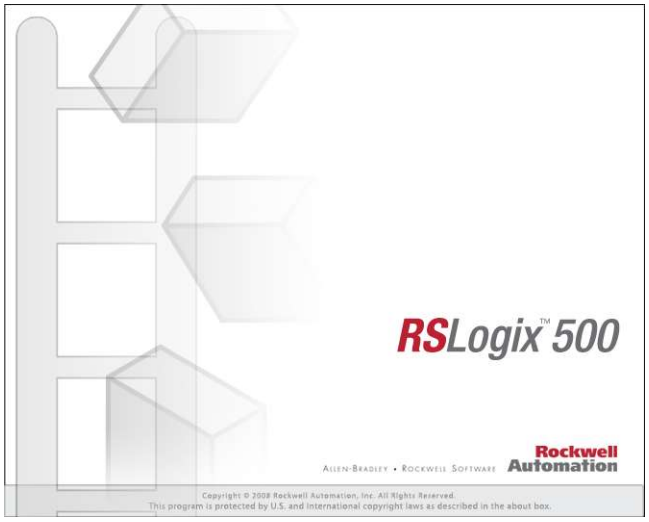
RSLogix Micro Project Report



Processor Information

Processor Type: Bul.1763 MicroLogix 1100 Series B

Processor Name: UNTITLED

Total Memory Used: 261 Instruction Words Used - 80 Data Table Words Used

Total Memory Left: 6395 Instruction Words Left

Program Files: 7

Data Files: 9

Program ID: 718c

I/O Configuration

0	Bul.1763	MicroLogix 1100 Series B
1	1762-IF2OF2	Analog 2 Chan. Input, 2 Chan. Output
2		
3		
4		

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

Source ID: 1 (decimal)
Baud: 19200
Parity: NONE
Control Line : No Handshaking
Error Detection: CRC
Embedded Responses: Auto Detect
Duplicate Packet Detect: Yes
ACK Timeout(x20 ms): 50
NAK Retries: 3
ENQ Retries: 3

CHANNEL 1 (SYSTEM) - Driver: Ethernet
CHANNEL 1 (SYSTEM) - Driver: Ethernet Edit Resource/Owner Timeout: 60
CHANNEL 1 (SYSTEM) - Driver: Ethernet Passthru Link ID: 1
CHANNEL 1 (SYSTEM) - Driver: Ethernet Write Protected: No
CHANNEL 1 (SYSTEM) - Driver: Ethernet Comms Servicing Selection: Yes
CHANNEL 1 (SYSTEM) - Driver: Ethernet Message Servicing Selection: Yes

Hardware Address: 00:00:00:00:00:00
IP Address: 0.0.0.0
Subnet Mask: 0.0.0.0
Gateway Address: 0.0.0.0
Msg Connection Timeout (x 1mS): 15000
Msg Reply Timeout (x mS): 3000
Inactivity Timeout (x Min): 30
Bootp Enable: Yes
Dhcp Enable No
SNMP Enable: No
HTTP Enable: Yes
Auto Negotiate Enable: Yes
Port Speed Enable: 10/100 Mbps Full Duplex/Half Duplex
Contact:
Location:

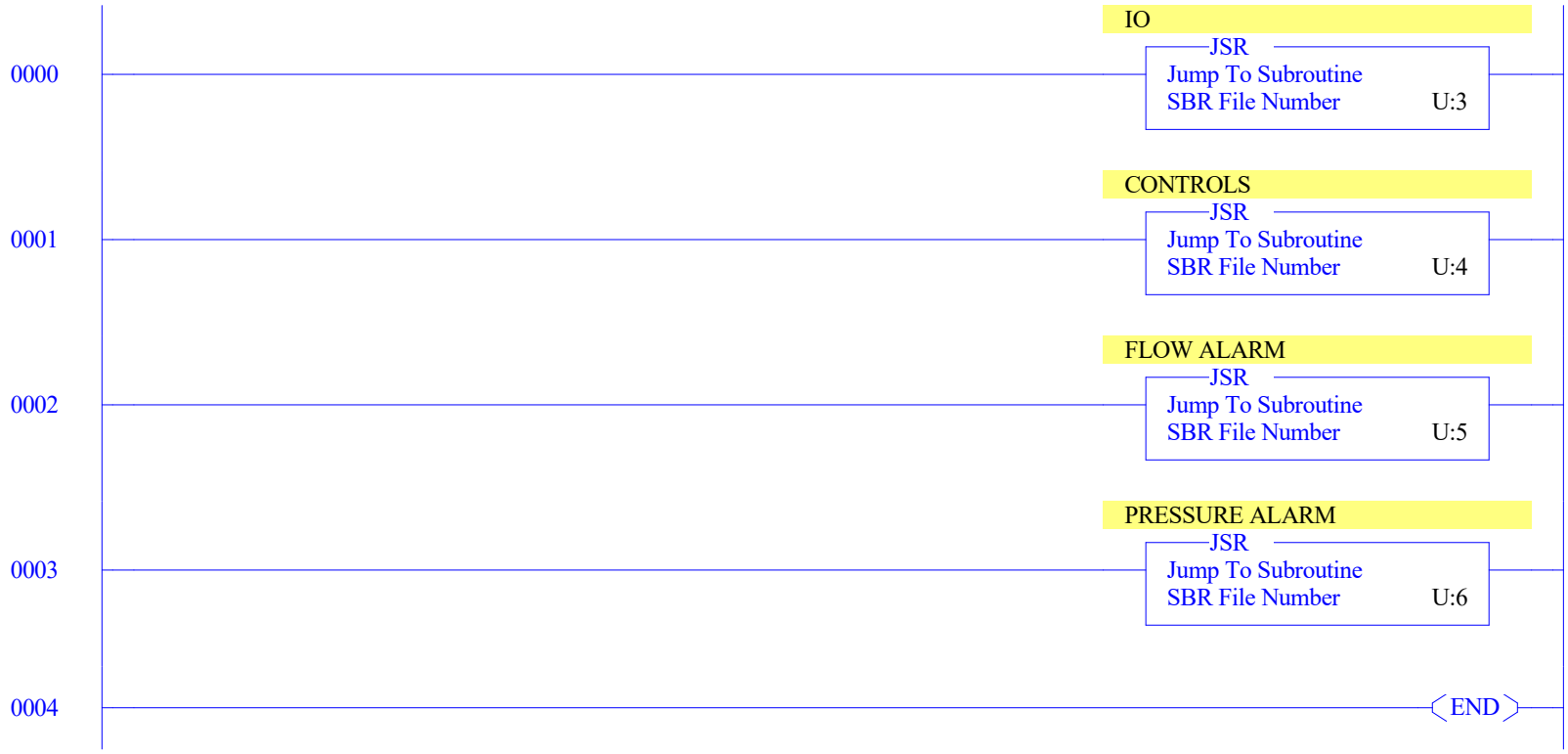
Program File List

Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
MAIN	2	LADDER	5	No	39
IO	3	LADDER	8	No	138
CONTROLS	4	LADDER	11	No	389
FLOW ALARM	5	LADDER	4	No	144
PRES ALARM	6	LADDER	4	No	150

Data File List

Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	18	6	O:5
INPUT	1	I	Global	No	36	12	I:11
STATUS	2	S	Global	No	0	66	S:65
BINARY	3	B	Global	No	2	2	B3:1
TIMER	4	T	Global	No	12	4	T4:3
COUNTER	5	C	Global	No	3	1	C5:0
CONTROL	6	R	Global	No	3	1	R6:0
INTEGER	7	N	Global	No	4	4	N7:3
FLOAT	8	F	Global	No	2	1	F8:0

LAD 2 - MAIN --- Total Rungs in File = 5



0000
flow switch input
I:0
0
Bul.1763
flow switch
B3:0
3

0001
alarm reset ip
I:0
1
Bul.1763
alarm reset
B3:0
4

0002
alarm silence ip
I:0
2
Bul.1763
alarm silence
B3:0
5

Scaling the signal coming from the pressure sensor

0003
pressure
SCP
Scale w/Parameters
Input I:1.0
0<
Input Min. 0
0<
Input Max. 16383
16383<
Scaled Min. 0
0<
Scaled Max. 55
55<
Output N7:1
27<

0004
pump energize
B3:0
6
Pump
O:0
0
Bul.1763

0005
flow fault indicato energize
B3:0
7
flow fault indicator
O:0
1
Bul.1763

0006
pressure fault indicator energize
B3:0
8
pressure fault indicator
O:0
2
Bul.1763

0007
END

The pump will go into "HAND" mode when the hand pushbutton is pressed. On pressing the button, the integer representing the mode the pump was in is temporarily moved to N7:3.

0000

hand pb B3:0 0 — flow alarm B3:1 4 — pressure alarm B3:1 8 — B3:1 ONS 14

intermediate mode bit

MOV
Move
Source N7:2 0<
Dest N7:3 0<

STATE
0=OFF
1=ON
2=AUTO

MOV
Move
Source 1 1<
Dest N7:2 0<

When the hand button is released, the pump will go back into the mode it was in before pressing the hand pushbutton.

0001

hand pb B3:0 0

OSF
One Shot Falling
Storage Bit B3:0/9
Output Bit B3:0/10

0002

revert back to original state B3:0 10

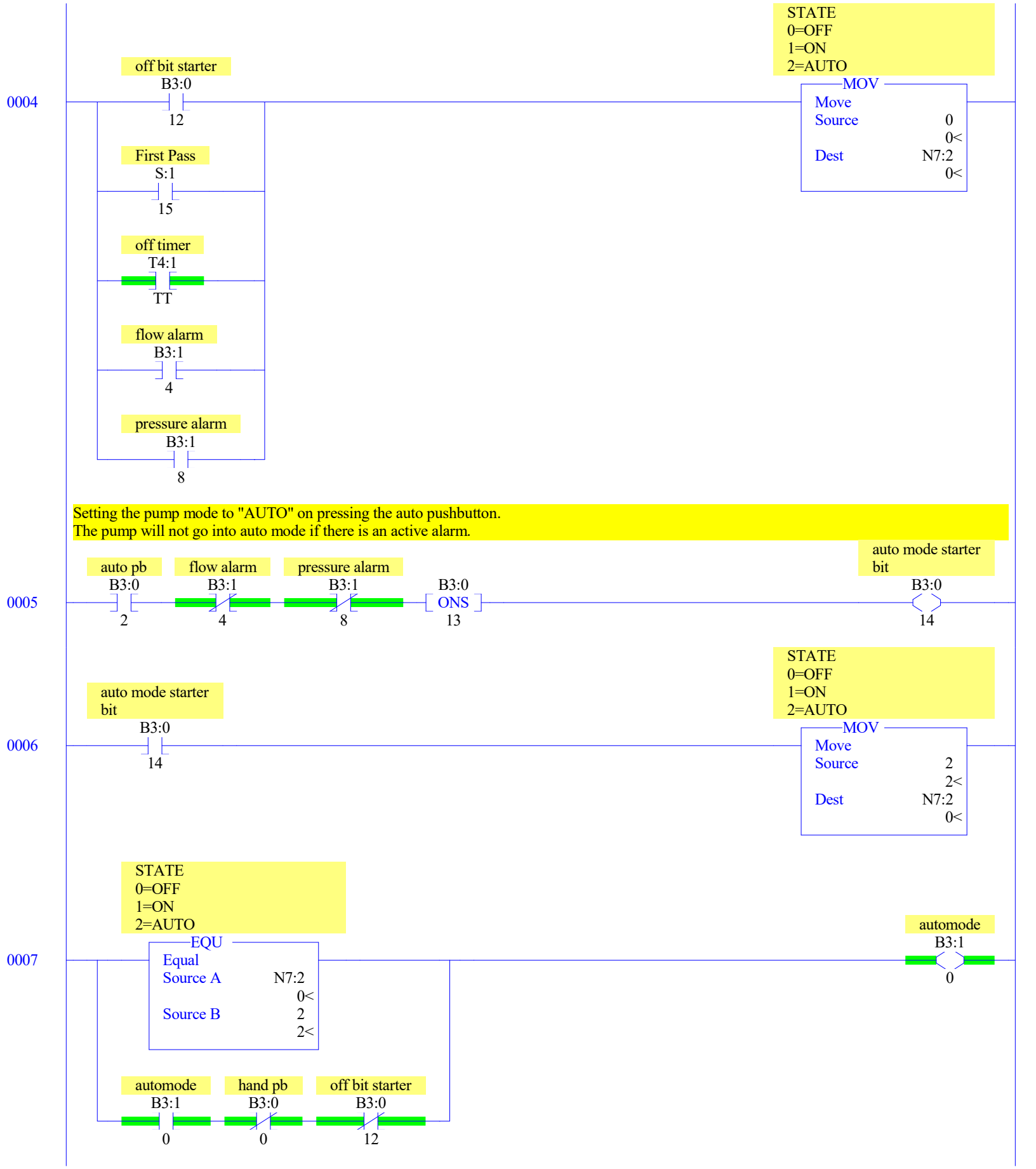
STATE
0=OFF
1=ON
2=AUTO

MOV
Move
Source N7:3 0<
Dest N7:2 0<

The pump goes into "OFF" mode when the off pushbutton is pressed, or when the PLC is first srted(first pass), or when any alarm is active.

0003

off pb B3:0 1 — B3:0 ONS 11 — off bit starter B3:0 12


0004
off bit starter
B3:0
12
First Pass
S:1
15
off timer
T4:1
TT
flow alarm
B3:1
4
pressure alarm
B3:1
8
STATE
0=OFF
1=ON
2=AUTO
MOV
Move
Source 0
0<
Dest N7:2
0<
Setting the pump mode to "AUTO" on pressing the auto pushbutton.
The pump will not go into auto mode if there is an active alarm.
0005
auto pb
B3:0
2
flow alarm
B3:1
4
pressure alarm
B3:1
8
B3:0
ONS
13
auto mode starter bit
B3:0
14
0006
auto mode starter bit
B3:0
14
STATE
0=OFF
1=ON
2=AUTO
MOV
Move
Source 2
2<
Dest N7:2
0<
0007
STATE
0=OFF
1=ON
2=AUTO
EQU
Equal
Source A N7:2
0<
Source B 2
2<
automode
B3:1
0
automode
B3:1
0
hand pb
B3:0
0
off bit starter
B3:0
12

When the pump is in "AUTO" mode, the pump will turn on for 30 seconds, then off for 10 seconds, and continue cycling this way, on and off, while it is in "AUTO" mode.

0008

automode
B3:1
0

pump on timer
TON
Timer On Delay
Timer T4:0
Time Base 1.0
Preset 30<
Accum 30<
EN
DN

pump on timer
T4:0
DN

off timer
TON
Timer On Delay
Timer T4:1
Time Base 1.0
Preset 10<
Accum 1<
EN
DN

off timer
T4:1
DN

pump on timer
T4:0
RES

0009

pump on timer
T4:0
TT

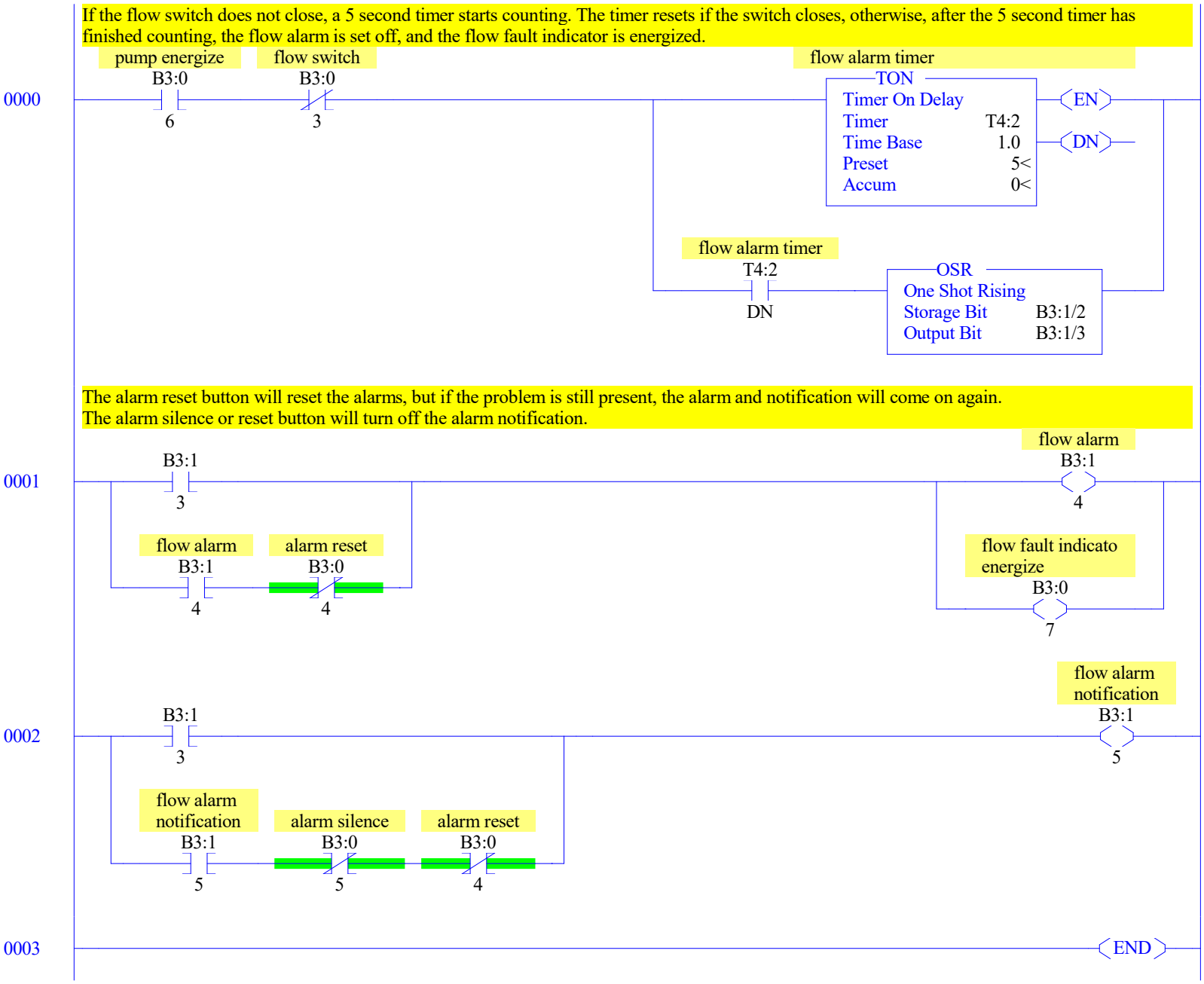
STATE
0=OFF
1=ON
2=AUTO

EQU
Equal
Source A N7:2
0<
Source B 1
1<

pump energize
B3:0
6

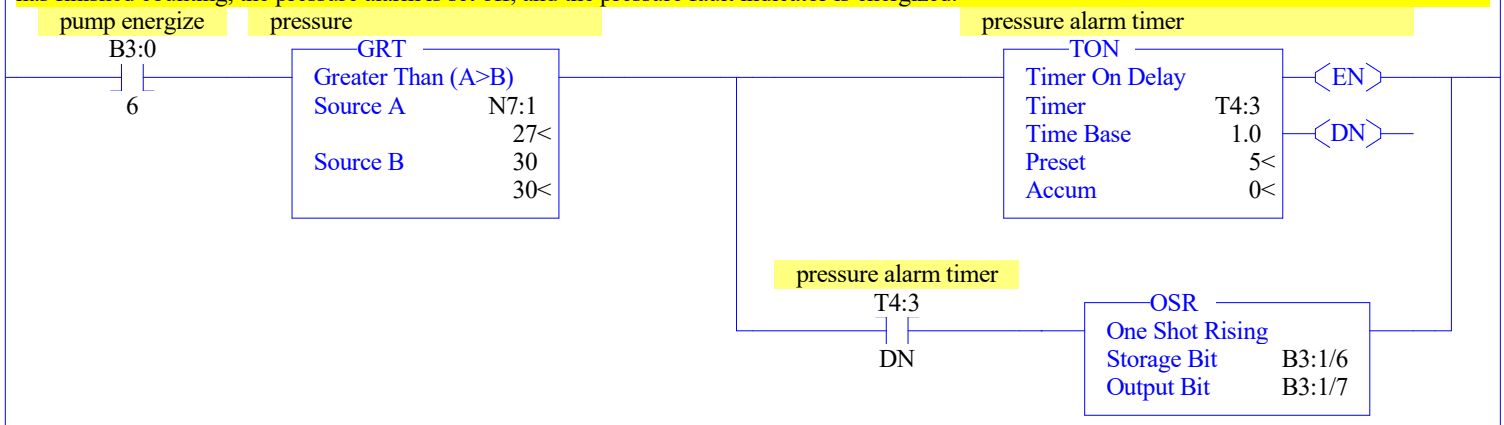
0010

END


If the flow switch does not close, a 5 second timer starts counting. The timer resets if the switch closes, otherwise, after the 5 second timer has finished counting, the flow alarm is set off, and the flow fault indicator is energized.
0000
pump energize
B3:0
6
flow switch
B3:0
3
flow alarm timer
TON
Timer On Delay
Timer T4:2
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
flow alarm timer
T4:2
DN
OSR
One Shot Rising
Storage Bit B3:1/2
Output Bit B3:1/3
The alarm reset button will reset the alarms, but if the problem is still present, the alarm and notification will come on again.
The alarm silence or reset button will turn off the alarm notification.
0001
B3:1
3
flow alarm
B3:1
4
flow alarm
B3:1
4
alarm reset
B3:0
4
flow fault indicato energize
B3:0
7
0002
B3:1
3
flow alarm notification
B3:1
5
flow alarm notification
B3:1
5
alarm silence
B3:0
5
alarm reset
B3:0
4
0003
END

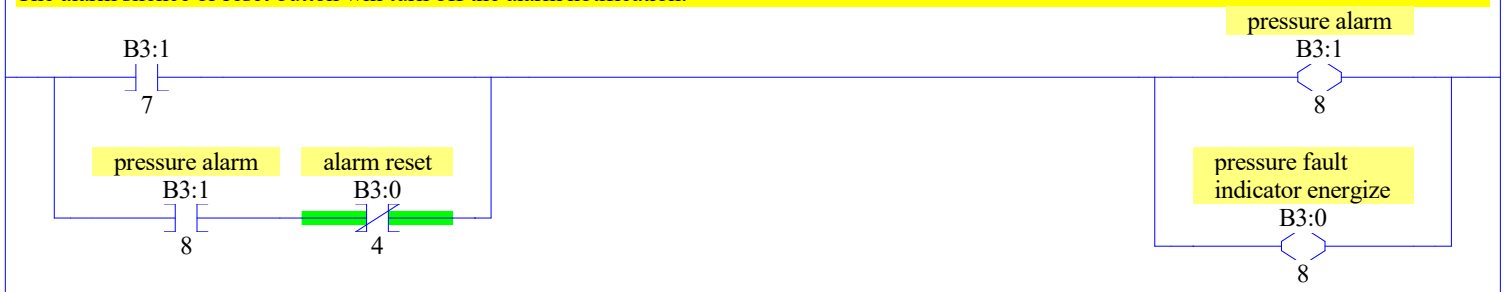
When the pressure exceeds 30 psi, a timer of 5 seconds starts. The timer resets if the pressure goes back below 30, otherwise, after the 5 second timer has finished counting, the pressure alarm is set off, and the pressure fault indicator is energized.

0000

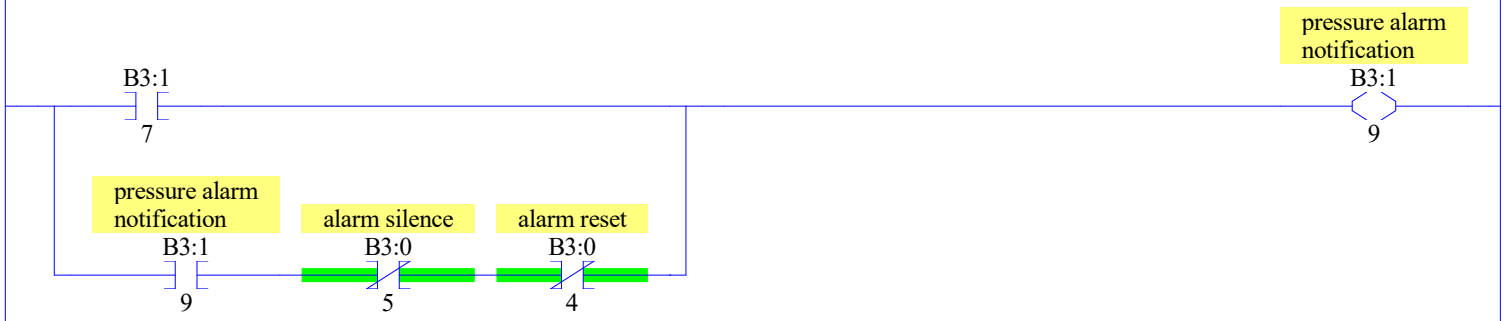


The alarm reset button will reset the alarms, but if the problem is still present, the alarm and notification will come on again.
The alarm silence or reset button will turn off the alarm notification.

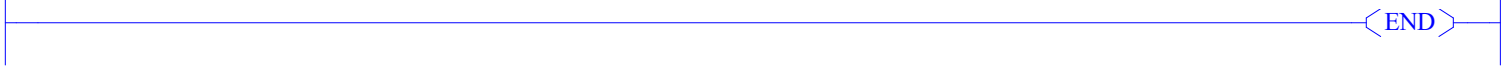
0001



0002



0003



Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
O:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
O:1.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
O:1.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
I:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	Bul.1763 MicroLogix 1100 Series B
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B
I:0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B-Analog
I:0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763 MicroLogix 1100 Series B-Analog
I:1.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
I:1.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
I:1.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
I:1.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
I:1.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O
I:1.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1762-IF2OF2 - Analog 2 Chan. Input, 2 Chan. O

Data File S2 (hex) -- STATUS

Main

Processor Mode S:1/0 - S:1/4 = Remote Run
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 0010-1010-0000-0100

Proc

OS Catalog Number S:57 = 1100
OS Series S:58 = B
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = A
Processor FRN S:62 =
User Program Type S:63 = 8001h
Compiler Revision Number S:64 =

Scan Times

Maximum (x10 ms) S:22 = 25
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 0
Scan Toggle Bit S:33/9 = 0

Math

Math Overflow Selected S:2/14 = 0
Overflow Trap S:5/0 = 0
Carry S:0/0 = 0
Overflow S:0/1 = 0
Zero Bit S:0/2 = 1
Sign Bit S:0/3 = 0
Math Register (lo word) S:13 = 0
Math Register (high word) S:14-S:13 = 0
Math Register (32 Bit) S:14-S:13 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = Remote Run
Node Address S:15 (low byte) = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0
Outgoing Msg Cmd Pending S:33/2 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0
Startup Protection Fault S:1/9 = 0
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0
Control Register Error S:5/2 = 0
Major Error Executing User Fault Rtn. S:5/3 = 0
Battery Low S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
ASCII String Manipulation error S:5/15 = 0
Fault Routine S:29 = 0
Major Error S:6 = 0h
Error Description:

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- BINARY

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B3:0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
B3:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol) Description
T4:0	1	0	1	1.0 sec	30	30	pump on timer
T4:1	1	1	0	1.0 sec	10	1	off timer
T4:2	0	0	0	1.0 sec	5	0	flow alarm timer
T4:3	0	0	0	1.0 sec	5	0	pressure alarm timer

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol)	Description
C5:0	0	0	0	0	0	0	0	0		

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol)	Description
R6:0	0	0	0	0	0	0	0	0	0	0		

Data File N7 (dec) -- INTEGER

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	8000	27	0	0						

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0				

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code	ABV	BLW
B3:0/0			hand pb				
B3:0/1			off pb				
B3:0/2			auto pb				
B3:0/3			flow switch				
B3:0/4			alarm reset				
B3:0/5			alarm silence				
B3:0/6			pump energize				
B3:0/7			flow fault indicato energize				
B3:0/8			pressure fault indicator energize				
B3:0/9							
B3:0/10			revert back to original state				
B3:0/11							
B3:0/12			off bit starter				
B3:0/13							
B3:0/14			auto mode starter bit				
B3:0/15			auto mode energizer				
B3:1/0			automode				
B3:1/1			handmode				
B3:1/2							
B3:1/3							
B3:1/4			flow alarm				
B3:1/5			flow alarm notification				
B3:1/6							
B3:1/7							
B3:1/8			pressure alarm				
B3:1/9			pressure alarm notification				
B3:1/14							
I:0/0			flow switch input				
I:0/1			alarm reset ip				
I:0/2			alarm silence ip				
I:1.0			pump pressure				
N7:0			prssure sensor input				
N7:1			pressure				
N7:2			STATE 0=OFF 1=ON 2=AUTO				
N7:3			intermediate mode bit				
O:0/0			Pump				
O:0/1			flow fault indicator				
O:0/2			pressure fault indicator				
S:0			Arithmetic Flags				
S:0/0			Processor Arithmetic Carry Flag				
S:0/1			Processor Arithmetic Underflow/ Overflow Flag				
S:0/2			Processor Arithmetic Zero Flag				
S:0/3			Processor Arithmetic Sign Flag				
S:1			Processor Mode Status/ Control				
S:1/0			Processor Mode Bit 0				
S:1/1			Processor Mode Bit 1				
S:1/2			Processor Mode Bit 2				
S:1/3			Processor Mode Bit 3				
S:1/4			Processor Mode Bit 4				
S:1/5			Forces Enabled				
S:1/6			Forces Present				
S:1/7			Comms Active				
S:1/8			Fault Override at Powerup				
S:1/9			Startup Protection Fault				
S:1/10			Load Memory Module on Memory Error				
S:1/11			Load Memory Module Always				
S:1/12			Load Memory Module and RUN				
S:1/13			Major Error Halted				
S:1/14			Access Denied				
S:1/15			First Pass				
S:2/0			STI Pending				
S:2/1			STI Enabled				
S:2/2			STI Executing				
S:2/3			Index Addressing File Range				
S:2/4			Saved with Debug Single Step				
S:2/5			DH-485 Incoming Command Pending				
S:2/6			DH-485 Message Reply Pending				
S:2/7			DH-485 Outgoing Message Command Pending				
S:2/15			Comms Servicing Selection				
S:3			Current Scan Time/ Watchdog Scan Time				
S:4			Time Base				
S:5/0			Overflow Trap				
S:5/2			Control Register Error				
S:5/3			Major Err Detected Executing UserFault Routine				
S:5/4			M0-M1 Referenced on Disabled Slot				
S:5/8			Memory Module Boot				
S:5/9			Memory Module Password Mismatch				
S:5/10			STI Overflow				
S:5/11			Battery Low				
S:6			Major Error Fault Code				
S:7			Suspend Code				
S:8			Suspend File				
S:9			Active Nodes				
S:10			Active Nodes				

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code	ABV	BLW
S:11			I/O Slot Enables				
S:12			I/O Slot Enables				
S:13			Math Register				
S:14			Math Register				
S:15			Node Address/ Baud Rate				
S:16			Debug Single Step Rung				
S:17			Debug Single Step File				
S:18			Debug Single Step Breakpoint Rung				
S:19			Debug Single Step Breakpoint File				
S:20			Debug Fault/ Powerdown Rung				
S:21			Debug Fault/ Powerdown File				
S:22			Maximum Observed Scan Time				
S:23			Average Scan Time				
S:24			Index Register				
S:25			I/O Interrupt Pending				
S:26			I/O Interrupt Pending				
S:27			I/O Interrupt Enabled				
S:28			I/O Interrupt Enabled				
S:29			User Fault Routine File Number				
S:30			STI Setpoint				
S:31			STI File Number				
S:32			I/O Interrupt Executing				
S:33			Extended Proc Status Control Word				
S:33/0			Incoming Command Pending				
S:33/1			Message Reply Pending				
S:33/2			Outgoing Message Command Pending				
S:33/3			Selection Status User/DF1				
S:33/4			Communicat Active				
S:33/5			Communicat Servicing Selection				
S:33/6			Message Servicing Selection Channel 0				
S:33/7			Message Servicing Selection Channel 1				
S:33/8			Interrupt Latency Control Flag				
S:33/9			Scan Toggle Flag				
S:33/10			Discrete Input Interrupt Reconfigur Flag				
S:33/11			Online Edit Status				
S:33/12			Online Edit Status				
S:33/13			Scan Time Timebase Selection				
S:33/14			DTR Control Bit				
S:33/15			DTR Force Bit				
S:34			Pass-thru Disabled				
S:34/0			Pass-Thru Disabled Flag				
S:34/1			DH+ Active Node Table Enable Flag				
S:34/2			Floating Point Math Flag Disable,Fl				
S:35			Last 1 ms Scan Time				
S:36			Extended Minor Error Bits				
S:36/8			DII Lost				
S:36/9			STI Lost				
S:36/10			Memory Module Data File Overwrite Protection				
S:37			Clock Calendar Year				
S:38			Clock Calendar Month				
S:39			Clock Calendar Day				
S:40			Clock Calendar Hours				
S:41			Clock Calendar Minutes				
S:42			Clock Calendar Seconds				
S:43			STI Interrupt Time				
S:44			I/O Event Interrupt Time				
S:45			DII Interrupt Time				
S:46			Discrete Input Interrupt- File Number				
S:47			Discrete Input Interrupt- Slot Number				
S:48			Discrete Input Interrupt- Bit Mask				
S:49			Discrete Input Interrupt- Compare Value				
S:50			Processor Catalog Number				
S:51			Discrete Input Interrupt- Return Number				
S:52			Discrete Input Interrupt- Accumulat				
S:53			Reserved/ Clock Calendar Day of the Week				
S:55			Last DII Scan Time				
S:56			Maximum Observed DII Scan Time				
S:57			Operating System Catalog Number				
S:58			Operating System Series				
S:59			Operating System FRN				
S:61			Processor Series				
S:62			Processor Revision				
S:63			User Program Type				
S:64			User Program Functional Index				
S:65			User RAM Size				
S:66			Flash EEPROM Size				
S:67			Channel 0 Active Nodes				
S:68			Channel 0 Active Nodes				
S:69			Channel 0 Active Nodes				
S:70			Channel 0 Active Nodes				
S:71			Channel 0 Active Nodes				
S:72			Channel 0 Active Nodes				
S:73			Channel 0 Active Nodes				
S:74			Channel 0 Active Nodes				

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Code	ABV	BLW
S:75			Channel 0 Active Nodes				
S:76			Channel 0 Active Nodes				
S:77			Channel 0 Active Nodes				
S:78			Channel 0 Active Nodes				
S:79			Channel 0 Active Nodes				
S:80			Channel 0 Active Nodes				
S:81			Channel 0 Active Nodes				
S:82			Channel 0 Active Nodes				
S:83			DH+ Active Nodes				
S:84			DH+ Active Nodes				
S:85			DH+ Active Nodes				
S:86			DH+ Active Nodes				
T4:0			pump on timer				
T4:0/DN							
T4:0/TT							
T4:1			off timer				
T4:1/DN							
T4:1/TT							
T4:2			flow alarm timer				
T4:2/DN							
T4:3			pressure alarm timer				
T4:3/DN							
U:3			IO				
U:4			CONTROLS				
U:5			FLOW ALARM				
U:6			PRESSURE ALARM				

Instruction Comment Database

Address	Instruction	Description

Symbol Group Database

Group_Name	Description